March 5, 1987

Selected Material
Relative to the FAA Investigation of
Japan Air Lines flight 1628
Sighting of Unidentified Air Traffic
on November 17, 1986.

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Prepared by
FAA Public Affairs Office
AAL-5
Alaskan Region
701 C St., Box 14, Anchorage, AK, 99513

U.S. Department of Transportation
Federal Aviation Administration

News:

Office of Public Affairs
Alaskan Region
701 C Street, Box 14
Anchorage, Alaska 99513
(907) 271-5296

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FOR RELEASE ON
MARCH 5, 1987
#87-09

CONTACT: PAUL STEUCKE

FAA RELEASES DOCUMENTS ON REPORTED UFO SIGHTING LAST NOVEMBER

The Federal Aviation Administration today released documents relating to the reported sighting of an unidentified flying object (UFO) over Alaska by a Japan Air Lines flight crew on November 17, 1986, saying it was unable to confirm the event.

The material was issued by FAA's Regional Office in Anchorage, Alaska, and included transcripts of pilot-controller communications, interviews with controllers and the flight crew, radar plots and other data.

FAA's Regional Public Affairs Officer Paul Steucke pointed out that FAA normally does not investigate UFO sightings but pursued the JAL incident in its role as the operator of the air traffic control system. He said the agency's objective was to determine if there was an unreported aircraft in the vicinity of the JAL flight that could present a safety hazard.

As part of the inquiry, Steucke said, radar data of the JAL flight track was reviewed by FAA experts at the agency's Technical Center in Atlantic City, N.J., using identical equipment. They determined that a second radar target near the JAL flight at the time of the reported sighting was not another aircraft but rather a split radar return from the JAL Boeing 747.

Technically, this is known as an "uncorrelated primary and beacon target return." It means that the primary radar signal reflected off the aircraft's surface did not correlate exactly with the pulse emitted by the aircraft's radar beacon transponder. This phenomenon is not unusual and gives the impression of two separate radar targets.

Steucke also noted that FAA controllers who monitored the JAL aircraft said in their statements that they thought there might have been another aircraft because of the dual radar targets. However, a northbound United Air Lines jet that was diverted by controllers to intercept the JAL flight path did establish visual contact with that aircraft but the pilots saw nothing else.

The Nov. 17 UFO sighting was reported by JAL Captain Kenjyu Terauchi on a cargo flight over the polar cap from Iceland to Japan via Anchorage. Captain Terauchi said he had visual contact from approximately the U.S.-Canadian border to south of Fairbanks. On Jan. 11, 1987, Captain Terauchi also reported another sighting in the same general area as the first.

Steucke said FAA is satisfied that the safety of the air traffic control system was not compromised by the Nov. 17 incident and plans no further investigation of the circumstances.

# # #

AAL-5, FAA
February 4, 1987

SELECTED PORTIONS OF TRANSCRIPTION
CONCERNING THE INCIDENT INVOLVING JAPAN AIRLINES FLIGHT 1628
ON NOVEMBER 18, 1986, AT APPROXIMATELY 0218 UTC
(Universal Time Coordinated - November 17, 1986
Alaska Standard Time, 6:18 p.m.)

0219:15	JL1628	Anchorage Center, Japan Air sixteen twenty eight; ah do you have any traffic, ah seven o'clock above?
0219:32	R/D15	Japan Air sixteen twenty eight heavy; negative.
0219:36	JL1628	Ah, Japan Air sixteen twenty eight; roger and, ah we insight-ah-two traffic-ah, in front of us one mile, about.
0221:19	R/D15	Japan Air sixteen twenty eight heavy; Sir if your able to identify the type of aircraft, ah-and see if you can tell whether its military or civilian.
0221:35	JL1628	Ah, Japan Air sixteen twenty eight; we cannot identify ah, the type, ah but, ah we can see, ah navigation lights and ah, strobe lights.
0221:48	R/D15	Roger sir, say the color of the strobe and beacon lights?
0221:56	JL1628	The color is ah, - white and yellow, I think.
0223:37	R/D15	Ya, could you (ROCC) look ah, approximately forty miles south of Fort Yukon, there should be a code up there of one five-five-zero. Can you tell me you see a primary target about his position?
0225:02	R/D15	Japan Air sixteen twenty eight heavy; roger. Sir, I'm picking up a-ah, hit on the radar approximately five miles in trail of your six o'clock position, do you concur?
0225:12	JL1628	Ah negative, ah eleven o'clock, ah eight miles, ah same level over.
0225:43	ROCC	Okay, I've got your squawk. It looks like I am getting some surge, primary return, ah I don't know if it's erroneous or whatever but...

JL1628 = Japan Air Lines flight #1628
R15 = Controller position, Anchorage Air Route Traffic Control Center
D15 = Controller position, Anchorage Air Route Traffic Control Center
ROCC = (Military) Regional Operations Command Center

0225:50	R/D15	Negative, uhuh, it's not erroneous. I want you (ROCC) to keep a good track on there, and if you pick up a code, and verify that you do not have any aircraft operating in that area military.
0225:57	ROCC	That is affirm. We do not have anybody up there right now, ah. Can you give me the position of the primary your receiving?
0226:03	R/D15	Okay, I'm not. I'm ah, picking up a primary - approximately five zero miles, south, right up there - right in front of the ah, one five five zero code.
0226:18	ROCC	Okay, I've got him about his-ah, oh-it looks like about, ah-ten o'clock, at about that range, yes.
0226:25	R/D15	Alright keep an eye on that, and ah-see if-ah, any other military (unintelligible) in that area.
0227:53	ROCC	It is an unknown, okay, we've lost contact with it now.
0230:56	JL1628	It's ah, I think ah, very quite big ah, plane.
0231:48	R15	Are you (Fairbanks FAA tower) pickin up a primary target right with that aircraft?
0231:58	FBKS	Ah, no. It's (the JL1628) the only target I see there.
0232:04	R15	If you see a primary with it, keep your eye on it. And he should be at three five oh(zero)also.
0232:05	FBKS	Alright, very good, we're watching.
0232:07	JL1628	Japan Air sixteen twenty eight, ah request descent.
0232:20	JL1628	Japan Air sixteen twenty eight, request three one zero.
0232:25	R15	Japan Air sixteen twenty eight heavy, understand, requesting flight level three one zero.
0232:41	R15	Japan Air sixteen one er correction sixteen twenty eight heavy, descend at pilot discretion-maintain flight level three one zero.
0232:45	JL1628	Leaving three five zero to three one zero.
0232:58	R15	Japan Air sixteen twenty eight heavy, do you still have your traffic?
0233:00	JL1628	Still, ah, coming ah, ah, right formation, in ah formation.

0234:52	R15	Japan Air sixteen twenty eight heavy, understand your traffic is over Fairbanks at this time.
0234:56	JL1628	Affirmative - - ah request heading two one zero.
0235:02	R15	Japan Air sixteen twenty eight heavy, roger. Deviations approved as necessary for traffic.
0235:24	D15	Approach center (Fairbanks FAA tower) on the sixty one line.
0235:27	FBKS	Approach.
0235:30	D15	Look at your radar. We got a Japan Air one six two eight, ah-he's deviating. He's about ah-five miles north of Fairbanks V-O-R squawking one five five zero.
0235:39	D15	Do you have any traffic with him? He sees traffic, he was deviating.
0235:44	FBKS	Ah - ya, no we don't. I don't see anything there aside from his, ah-his target.
0236:12	JL1628	Ah-Anchorage Center; Japan Air one six two eight, request direct ah-Talkeetna.
0236:18	R15	Japan Air sixteen twenty eight heavy, cleared direct Talkeetna, and in-ah-advise me of your position of your traffic?
0236:24	JL1628	Ah, same po, same position.
0236:37	R15	Japan Air sixteen twenty eight heavy, ah-sir I'm gonna request you to make a right turn, three six zero degrees, three hundred and sixty degree turn, and advise me what your traffic does then.
0236:47	JL1628	Right turn, three sixty.
0237:23	ROCC	Okay. We have, no, we have confirmed, we have no military aircraft working up there.
0237:25	D15	Okay thank you very much.
0237:29	D15	You have no traffic at all?
0237:30	ROCC	That's correct, does he (JAL-1628) still have somebody visual?
0237:32	D15	He says he does.

FBKS = Fairbanks Airport FAA Approach Control (Tower)

0237:55	R15	Japan Air sixteen twenty eight heavy. Sir, does your traffic appear to be staying with you?
0238:00	JL1628	(unintelligible) just looking.
0239:01	R15	Japan Air sixteen twenty eight. Say again?
0239:04	JL1628	It ah-disappeared. Japan Air sixteen twenty eight.
0239:58	ROCC	Ya, this is one dash two again. On some other equipment here we have confirmed there is a flight size of two around your one five five zero. Squawk one primary return only.
0240:05	D15	Okay, where is is he following him?
0240:07	ROCC	It looks like he is, yes.
0240:10	R15	Japan Air sixteen twenty eight heavy; roger. At your discretion proceed direct Talkeetna, Jay one two five Anchorage. (J125 = air route)
0240:23	R15	Japan Air sixteen twenty eight; roger. Sir, the military radar advises they do have a primary target in trail of you at this time.
0240:24	D15	Okay do you have anybody you can scramble up there?
0240:30	ROCC	I'll tell you what, we're gonna talk to your liasion sir about that.
0240:35	R15	Japan Air sixteen twenty eight heavy. Military radar advises they are picking up intermittent primary target behind you in-trail, in-trail I say again.
0240:51	ROCC	Ah-I'm gonna talk to my other radar man here - has gotta, he's got some other equipment watching this aircraft.
0240:54	D15	Okay.
0240:13	R15	Roger sir. Would you (JAL 1628) like our military to scramble on the traffic?
0240:17	JL1628	Negative, negative.
0242:04	JL1628	Anchorage Center. Japan Air sixteen twenty eight. Confirm direct to Talkeetna, three one zero.
0242:09	R15	Japan Air sixteen twenty eight heavy; affirmative. Direct Talkeetna and descend at pilot's discretion, maintain flight level two five zero.

0242:24	ROCC	It looks like he, he-ah-offset left, and then possibly fell back in-trail. However, I can't see him now, I can't pick him out.
0242:29	D15	Okay thank you very much S-R.
0244:13	R15	Japan Air sixteen twenty eight heavy. Do you still have the traffic?
0244:17	JL1628	Ah-affirmative. Ah-nine o'clock, ah, ah.
0244:39	R15	United sixty nine, Anchorage request.
0244:41	UA69	Ah, go ahead, United sixty nine heavy.
0244:43	R15	United sixty nine heavy. Sir, I've got a Japan Air seven forty seven presently in your eleven o'clock position and one hundred and one, correction, one, one, zero miles, and he has traffic (unintelligible) I'll keep you advised, ah-when you're closer to him I want you to see if you see anything with him.
0245:04	UA69	Okay fine, ah-we'll look for ya.
0245:49	R15	United sixty nine heavy. In your eleven o'clock position, one, zero, zero miles, southbound, is a Japan Air, seven forty seven. He is at flight level three one zero. Says he has traffic at his nine o'clock position same altitude.
0246:06	UA69	We'll be looking, that's ah-ah, can't see anything yet.
0246:59	R15	Japan Air sixteen twenty eight; roger. I'm gonna have a United aircraft get close to you and take a look, ah-to see if he can identify your traffic.
0247:06	JL1628	Thank you.
0247:09	R15	United sixty nine heavy, turn ten degrees left, radar vectors, ah-to see traffic.
0248:31	R15	Japan Air sixteen twenty eight heavy. Say the position of your traffic.
0248:34	JL1628	Ah-now, ah-ah-moving to ah-around ten mile, now-ah-ah-position-ah-seven, ah-eight o'clock, ten mile.
0248:52	R15	Japan Air sixteen seventy eight heavy; roger.

UA69 = United Airlines flight #69

0249:52	R15	United sixty nine heavy; roger. The Japan Air says the traffic is in his seven to eight o'clock position and one zero miles in-trail.
0250:00	UA69	Okay, we're lookin.
0250:05	UA69	Why don't you get us a little closer?
0250:07	R15	United sixty nine heavy; roger. Another additional ten degrees left.
0250:14	UA69	Roger, United sixty nine.
0250:46	R15	United sixty nine, that's what he says. Japan Air sixteen twenty eight heavy, say the position of your traffic now.
0250:52	JL1628	Ah-now distinguishing but, ah-ah-your I guess, ah-twelve o'clock below-ah-you, over.
0251:32	UA69	Ah, Center from United ah sixty nine. Ah-the-ah-Japan Airliner is silhouetted against a-ah-light sky. I don't see anybody around him at all. I can see his contrail but I sure don't see any other airplanes. Do you see him?
0251:49	R15	United sixty nine heavy, ah-negative sir. We got just a very few primary hits on the ah-target and then ah-we really haven't got a good track on him ever.
0252:31	TOTEM	If you want we've got extra gas we could bop up another five or six thousand feet and turn around.
0252:36	R15	Totem (military C130 flight) seven one, ah roger sir. If you'd like ah-standby, ---- Totem seven one, turn ah-right, heading two five, correction turn right, heading two seven zero, radar vectors to intercept.
0253:10	R15	Japan Air sixteen twenty eight heavy, descend at pilot discretion, maintain flight level two five zero.
0253:13	JL1628	Japan Air sixteen twenty eight, ah-pilot's discretion, maintain ah-ah-two five zero, so-ah-ah-I cannot, I couldn't see ah-U-F-O, over.
0253:27	R15	Japan Air sixteen twenty eight heavy. Understand you do not see the traffic any longer.
0253:31	JL1628	Affirmative.
0254:04	TOTEM	Okay sir, we're searching this time, we think we have him.

TOTEM = Military C130 aircraft

0254:09	R15	Japan Air sixteen twenty eight heavy, flash your landing lights please.
0255:25	TOTEM	Ah, yes sir, we've got him insight.
0255:35	R15	Totem seven one, do you see any traffic in his vicinity?
0255:38	TOTEM	Not flashing any lights at this time sir.
0323:		JL1628 landed at Anchorage International Airport (approximate)

ANALYSIS OF UNCORRELATED PRIMARY
AND BEACON TARGETS
(JAL-1628, 11/17/86 AKST)

Approximately 61 minutes of data was extracted from the EARTS CDR printouts relating to the November 18 incident involving JAL-1628 and the alleged UFO sighting.

Review of the data involving this incident did not show any abnormalities that could be associated with any type of target as indicated by the pilot of JAL-1628.

Radar returns from the aircraft and surrounding terrain vary with the different segments of the flight, but are considered normal for the area.

Returns relating to the incident can be categorized as three types: primary radar reinforced by a beacon reply (primary radar returns and beacon returns are both evident in the same 1/4 mile range cell), beacon only reply and beacon with an associated radar reply. Seventy-two percent of the replies were radar with beacon reinforcement (same range cell) which is normal for the Murphy dome radar system.

Approximately 25 percent were beacon only and of those that registered as beacon only, 90 percent of those had a primary only reply within 1/8 of a mile, either ahead or behind the beacon target (5 behind, 12 ahead).

These uncorrelated primary returns are not uncommon, due to the critical timing associated with the delay adjustments in the aircraft transponder for beacon systems and the target correlation circuitry within the radar equipment.

When an aircraft is being interrogated as it passes through the beginning of adjacent range cells the intricate timing between the two systems very often is off just enough to declare both a beacon and a radar target in different range cells, resulting in uncorrelated radar replies.

The data derived from the JAL-1628 flight is representative of the data from another aircraft in the same general area and is considered normal.

February 25, 1987

DS

Dennis R. Simantel
ZAN-AAL-ARTCC

CONCUR: David F. Morse, AAL-400